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IPV6

- The Internet Protocol version 6, or IPv6, is the latest version of the Internet Protocol (IP), which is the system used for identifying and locating computers on the Internet.
- IPv6 was developed by the Internet Engineering Task Force (IETF) to deal with the problem of IPv4 exhaustion.
- IPv6 is a **128-bit** address having an address space of $\mathbf{2^{128}}$, which is way bigger than IPv4. IPv6 uses a hexadecimal format separated by a colon **(:)**.

Components in IPv6 Address Format

- There are 8 groups, and each group represents 2 Bytes (16 bits).
- Each Hex-Digit is of 4 bits (1 nibble)
- Delimiter used - colon (:)

Addressing

There are 3 types of addresses:

- unicast, anycast, and multicast
- Approximately **15%** of the address space is initially allocated to the reserved addresses, such as **NSAP** (Network Access Service Point) addresses, **IPX** (Internetworking Packet Exchange) addresses, etc.
- The remaining 85% is reserved for future use.

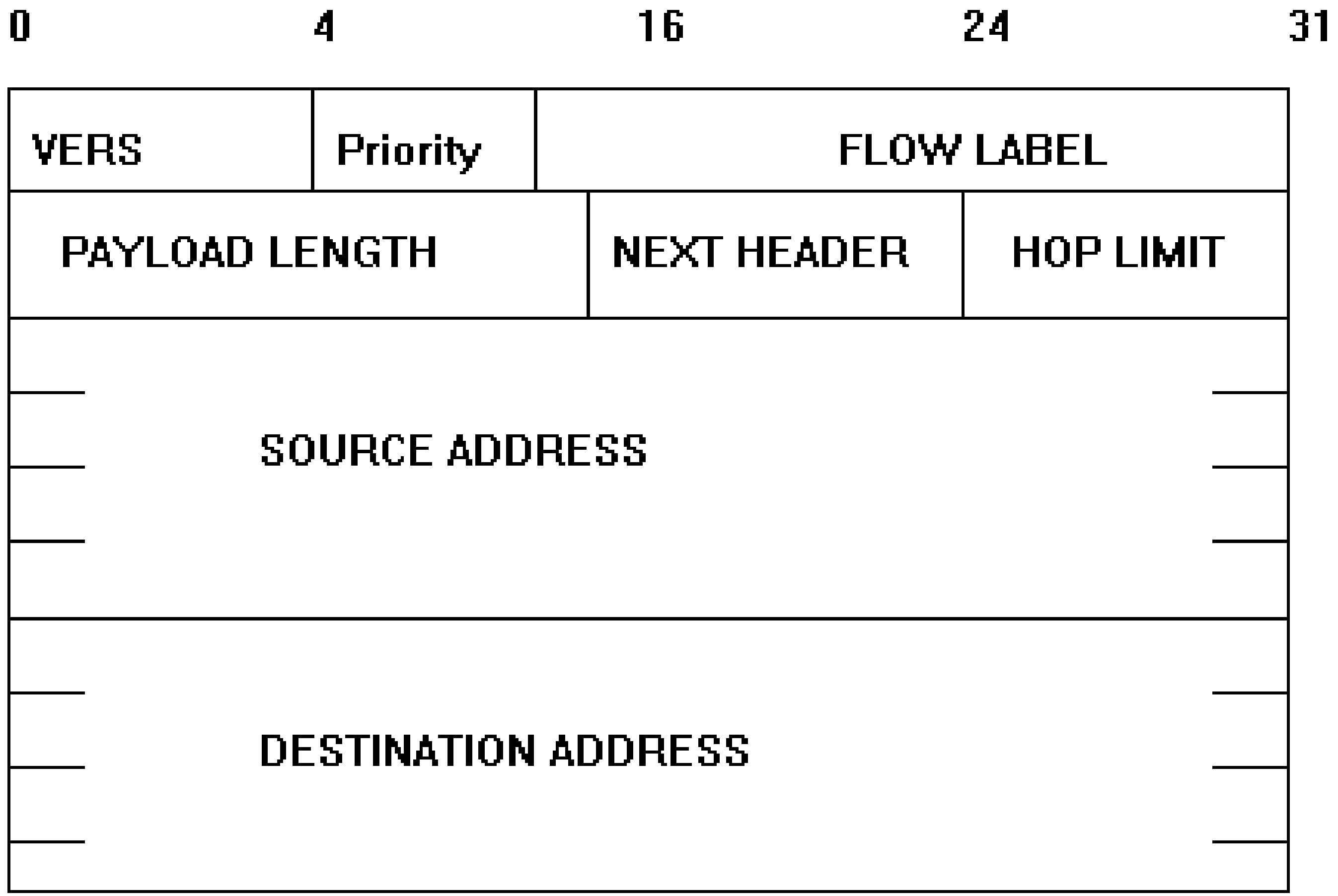
- **Unicast Address**: Unicast Address identifies a single network interface. A packet sent to a unicast address is delivered to the interface identified by that address.

- **Multicast Address**: A Multicast Address is used by multiple hosts, called groups, to acquire a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address. And every node is configured in the same way. In simple words, one data packet is sent to multiple destinations simultaneously.

- **Anycast Address*:*** An Anycast Address is assigned to a group of interfaces. Any packet sent to an anycast address will be delivered to only one member interface (mostly the nearest host possible).

- Broadcast is not defined in IPv6.

Header Format



VERS
identifies protocol as version 6

PRIORITY
Identifies Priority characteristics.

PAYLOAD LENGTH
Specifies only the size of the data being carried; it does not include the header.

HOP LIMIT
Corresponds to IP's TIME TO LIVE; the datagram is discarded if the HOP LIMIT counts down to zero.

FLOW LABEL
Divided into two parts, one used to define a specific path, the other to specify a traffic class. Routers use the value in the FLOW LABEL field to route the datagram.

NEXT HEADER
Used to specify the type of information that follows the current header.